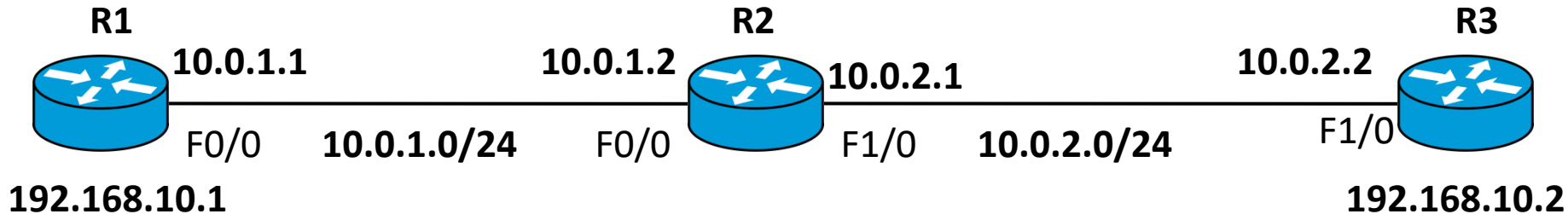


EIGRP LAB

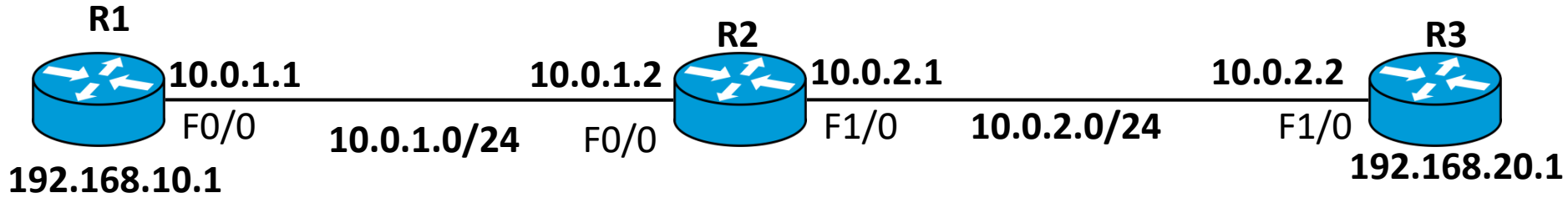


Explanation:

R1 is connected to R2 and R2 is connected to R3
Where R1 is having Loopback IP 192.168.10.1
And R2 is having Loopback IP 192.168.10.1

Lets First Provide IP Address to All these.

Step 1: Provide IP: R1



```
R1>enable
```

```
R1# configure terminal
```

```
R1(config)#interface loopback 1
```

```
R1(config-if)#ip address 192.168.10.1 255.255.255.0
```

```
R1(config-if)#exit
```

```
R1(config)#int f0/0
```

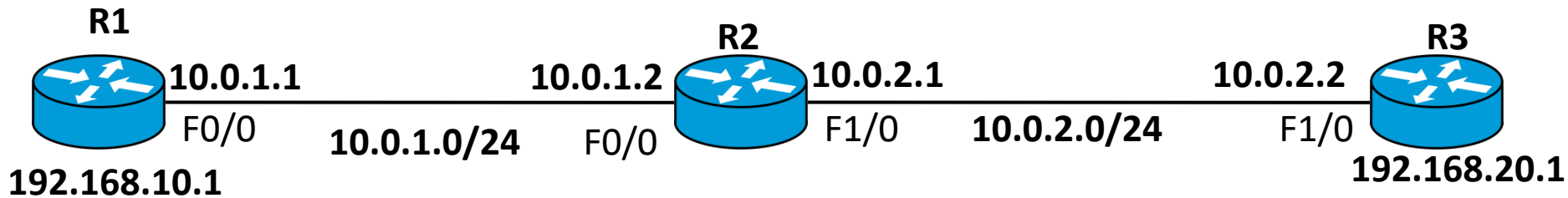
```
R1(config-if)#no sh
```

```
R1(config-if)#no shutdown
```

```
R1(config-if)#ip address 10.0.1.1 255.255.255.0
```

```
R1(config-if)#exit
```

Step 1: Provide IP: R2



```
R2>enable
```

```
R2# configure terminal
```

```
R2(config)#int f0/0
```

```
R2(config-if)#no shutdown
```

```
R2(config-if)#ip address 10.0.1.2 255.255.255.0
```

```
R2(config-if)#exit
```

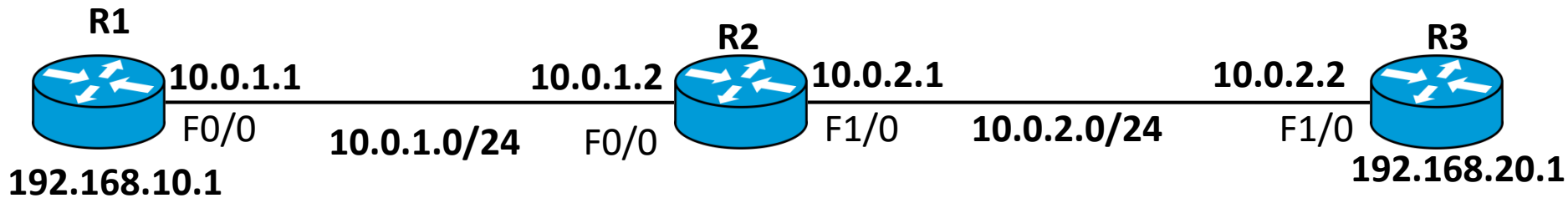
```
R2(config)#int f1/0
```

```
R2(config-if)#no shutdown
```

```
R2(config-if)#ip address 10.0.2.1 255.255.255.0
```

```
R2(config-if)#exit
```

Step 1: Provide IP: R3



```
R3>enable
```

```
R3# configure terminal
```

```
R3(config)#interface loopback 1
```

```
R3(config-if)#ip address 192.168.20.1 255.255.255.0
```

```
R3(config-if)#exit
```

```
R3(config)#int f1/0
```

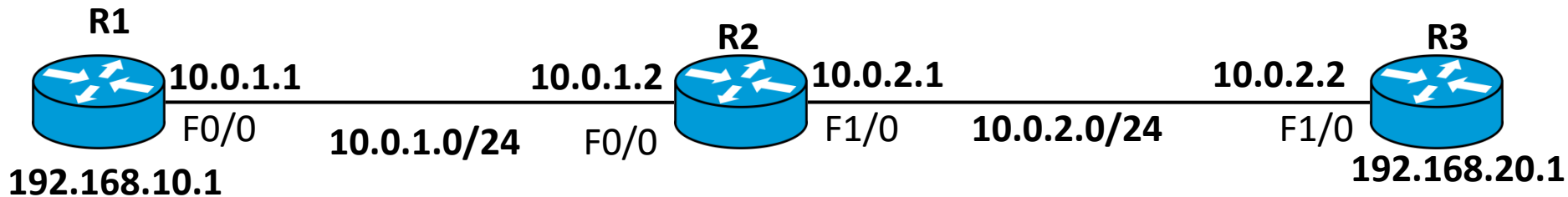
```
R3(config-if)#no sh
```

```
R3(config-if)#no shutdown
```

```
R3(config-if)#ip address 10.0.2.2 255.255.255.0
```

```
R3(config-if)#exit
```

Step 2: Configure EIGRP on R1



```
R1>enable
```

```
R1# configure terminal
```

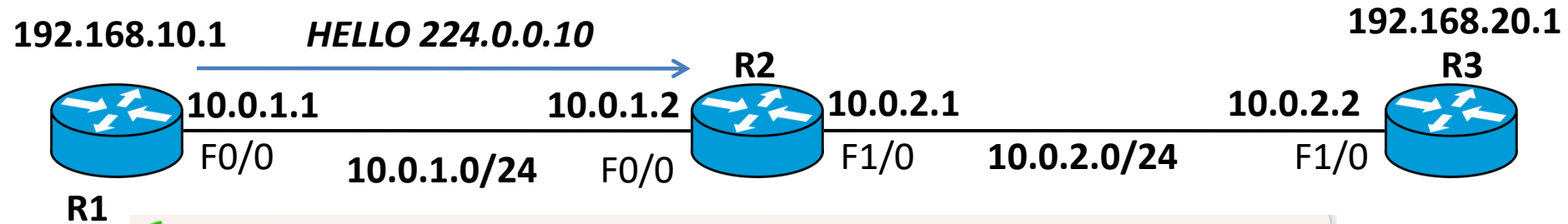
```
R1(config)#router eigrp 1
```

```
R1(config-router)# network 192.168.10.0
```

```
R1(config-router)# network 10.0.1.0
```

```
R1(config)# exit
```

Hello packet will Start Multicast From R1



Capturing from - [R1 FastEthernet0/0 to R2 FastEthernet0/0]

EIGRP PACKET

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

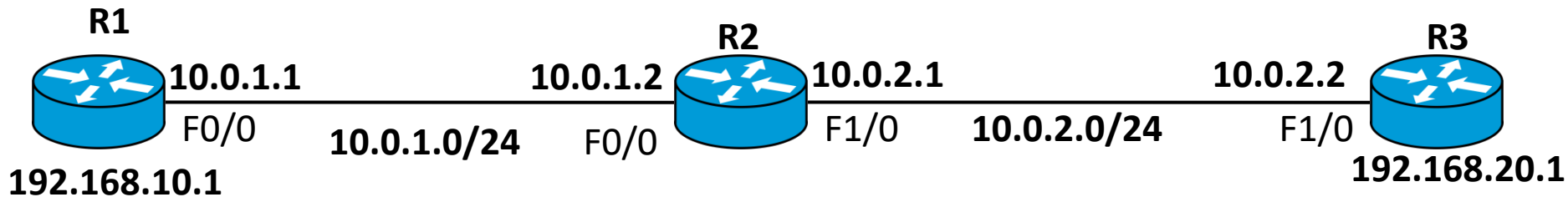
No.	Time	Source	Destination	Protocol	Length	Info
7	20.043893	ca:01:3e:2c:00:00	ca:01:3e:2c:00:00	LOOP	60	Reply
8	23.247605	10.0.1.1	224.0.0.10	EIGRP	74	Hello
9	27.839599	10.0.1.1	224.0.0.10	EIGRP	74	Hello
10	29.998841	ca:02:0c:30:00:00	ca:02:0c:30:00:00	LOOP	60	Reply
11	30.046501	ca:01:3e:2c:00:00	ca:01:3e:2c:00:00	LOOP	60	Reply
12	32.654479	10.0.1.1	224.0.0.10	EIGRP	74	Hello
13	36.164364	ca:01:3e:2c:00:00	CDP/VTP/DTP/PAgP/UDLD	CDP	366	Device ID: R1 Port ID
14	37.263375	10.0.1.1	224.0.0.10	EIGRP	74	Hello
15	39.994319	ca:02:0c:30:00:00	ca:02:0c:30:00:00	LOOP	60	Reply
16	40.041194	ca:01:3e:2c:00:00	ca:01:3e:2c:00:00	LOOP	60	Reply
17	41.641412	10.0.1.1	224.0.0.10	EIGRP	74	Hello
18	46.004274	10.0.1.1	224.0.0.10	EIGRP	74	Hello
19	49.991189	ca:02:0c:30:00:00	ca:02:0c:30:00:00	LOOP	60	Reply

Ready to load or capture

Packets: 128 · Displayed: 128 (100.0%) · Filter: Default

HELLO
PACKET
WILL
START
MULTICAST
FROM R1

Step 2: Configure EIGRP on R2



```
R2>enable
```

```
R2# configure terminal
```

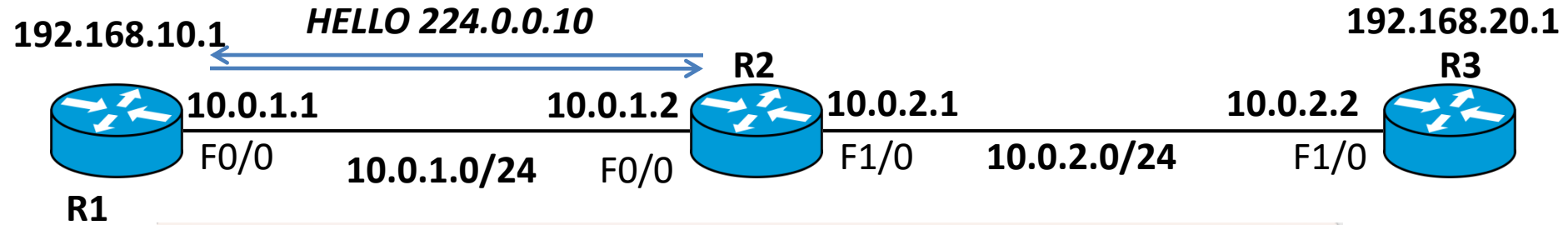
```
R2(config)#router eigrp 1
```

```
R2(config-router)# network 10.0.1.0
```

```
R2(config-router)# network 10.0.2.0
```

```
R2(config)# exit
```

Hello packet will Start Multicast From R1 and R2 Both



Capturing from - [R1 FastEthernet0/0 to R2 FastEthernet0/0] **EIGRP PACKET**

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

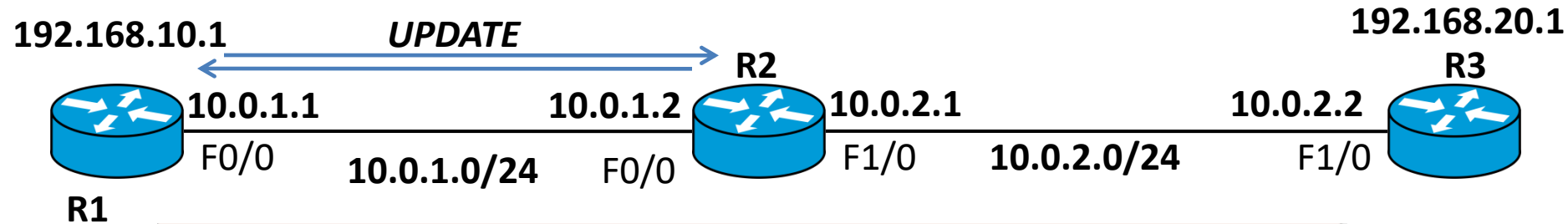
No.	Time	Source	Destination	Protocol	Length	Info
1706	3739.981210	ca:01:3e:2c:00:00	ca:01:3e:2c:00:00	LOOP	60	Reply
1707	3742.981940	10.0.1.1	224.0.0.10	EIGRP	74	Hello
1708	3744.066628	10.0.1.2	224.0.0.10	EIGRP	74	Hello
1709	3744.098627	10.0.1.1	224.0.0.10	EIGRP	84	Hello
1710	3744.098627	ca:01:3e:2c:00:00	Broadcast	ARP	60	Who has 10.0.1.2? Tell
1711	3744.129552	ca:02:0c:30:00:00	ca:01:3e:2c:00:00	ARP	60	10.0.1.2 is at ca:02:0
1712	3744.159725	10.0.1.2	224.0.0.10	EIGRP	84	Hello
1713	3744.175165	10.0.1.2	10.0.1.1	EIGRP	60	Update
1714	3746.105743	10.0.1.1	10.0.1.2	EIGRP	60	Update
1715	3746.135747	10.0.1.2	10.0.1.1	EIGRP	60	Hello (Ack)
1716	3746.166275	10.0.1.1	224.0.0.10	EIGRP	98	Update
1717	3746.182277	10.0.1.2	10.0.1.1	EIGRP	60	Hello (Ack)
1718	3746.198372	10.0.1.2	224.0.0.10	EIGRP	98	Update

Ready to load or capture

Packets: 1759 • Displayed: 1759 (100.0%) • Profile: Default

**HELLO
PACKET
WILL
START
MULTICAST
FROM
R1 and R2**

Once Create Neighborship, update will be forwarded:



Capturing from - [R1 FastEthernet0/0 to R2 FastEthernet0/0]

EIGRP PACKET

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

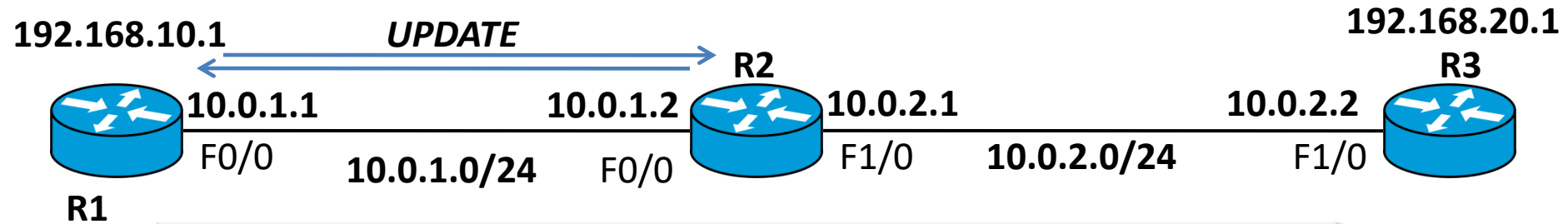
No.	Time	Source	Destination	Protocol	Length	Info
1706	3739.981210	ca:01:3e:2c:00:00	ca:01:3e:2c:00:00	LOOP	60	Reply
1707	3742.981940	10.0.1.1	224.0.0.10	EIGRP	74	Hello
1708	3744.066628	10.0.1.2	224.0.0.10	EIGRP	74	Hello
1709	3744.098627	10.0.1.1	224.0.0.10	EIGRP	84	Hello
1710	3744.098627	ca:01:3e:2c:00:00	Broadcast	ARP	60	Who has 10.0.1.2? Tell
1711	3744.129552	ca:02:0c:30:00:00	ca:01:3e:2c:00:00	ARP	60	10.0.1.2 is at ca:02:0
1712	3744.159725	10.0.1.2	224.0.0.10	EIGRP	84	Hello
1713	3744.175165	10.0.1.2	10.0.1.1	EIGRP	60	Update
1714	3746.105743	10.0.1.1	10.0.1.2	EIGRP	60	Update
1715	3746.135747	10.0.1.2	10.0.1.1	EIGRP	60	Hello (Ack)
1716	3746.166275	10.0.1.1	224.0.0.10	EIGRP	98	Update
1717	3746.182277	10.0.1.2	10.0.1.1	EIGRP	60	Hello (Ack)
1718	3746.198372	10.0.1.2	224.0.0.10	EIGRP	98	Update

Ready to load or capture

Packets: 1759 • Displayed: 1759 (100.0%) • Filter: Default

UPDATE
Will be
Forwarded
To each
other

After Update Acknowledgement will be forwarded:



Capturing from - [R1 FastEthernet0/0 to R2 FastEthernet0/0] **EIGRP PACKET** X

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

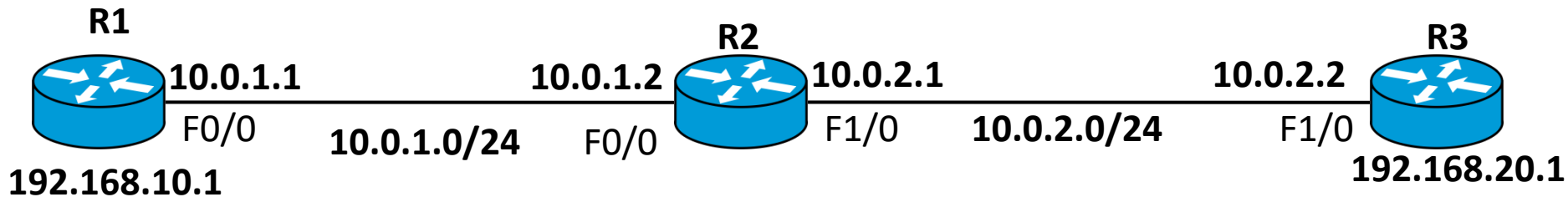
No.	Time	Source	Destination	Protocol	Length	Info
1706	3739.981210	ca:01:3e:2c:00:00	ca:01:3e:2c:00:00	LOOP	60	Reply
1707	3742.981940	10.0.1.1	224.0.0.10	EIGRP	74	Hello
1708	3744.066628	10.0.1.2	224.0.0.10	EIGRP	74	Hello
1709	3744.098627	10.0.1.1	224.0.0.10	EIGRP	84	Hello
1710	3744.098627	ca:01:3e:2c:00:00	Broadcast	ARP	60	Who has 10.0.1.2? Tell
1711	3744.129552	ca:02:0c:30:00:00	ca:01:3e:2c:00:00	ARP	60	10.0.1.2 is at ca:02:0
1712	3744.159725	10.0.1.2	224.0.0.10	EIGRP	84	Hello
1713	3744.175165	10.0.1.2	10.0.1.1	EIGRP	60	Update
1714	3746.105743	10.0.1.1	10.0.1.2	EIGRP	60	Update
1715	3746.135747	10.0.1.2	10.0.1.1	EIGRP	60	Hello (Ack)
1716	3746.166275	10.0.1.1	224.0.0.10	EIGRP	98	Update
1717	3746.182277	10.0.1.2	10.0.1.1	EIGRP	60	Hello (Ack)
1718	3746.198372	10.0.1.2	224.0.0.10	EIGRP	98	Update

Ready to load or capture

Packets: 1759 • Displayed: 1759 (100.0%) • Filter: Default

Acknowled-
gement
Will be
Forwarded
To each
other

Step 2: Configure EIGRP on R3



```
R3>enable
```

```
R3# configure terminal
```

```
R3(config)#router eigrp 1
```

```
R3(config-router)# network 192.168.20.0
```

```
R3(config-router)# network 10.0.2.0
```

```
R3(config)# exit
```